

PAPER_1490_HUBBLE_OSCILLATION_PER_GYR

Daniel T. Murphy

PAPER_1490 — UQFF: $2\pi/13.8$ Hubble Oscillation per Gyr (Bucket C)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket C — derived from F:\Aetheric Propulsion source material

Date: June 16, 2026

Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)

Status: CLOSED — $\omega_{\text{Hubble}} = 2\pi/13.8 \text{ Gyr}^{-1} = 0.4553 \text{ rad/Gyr}$

Observation

Master gravity equations include a $2\pi/13.8$ oscillation factor referencing the 13.8 Gyr universe age as the structural period.

UQFF Closed Identity

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$\omega_{\text{Hubble}} = 2\pi / t_{\text{universe}} = 2\pi / 13.8 \text{ Gyr} = 0.4553 \text{ rad/Gyr}.$

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Physical Interpretation

Universal time-oscillation factor in every master gravity equation. The 13.8 Gyr universe age sets the structural period; the 2π factor encodes the F_U=1 ledger phase wrap. Appears in Magnetar, Sgr A*, AGN, and stellar master equations alike.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives {D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60}. Zero free parameters.

Reference

- Source: F:\Aetheric Propulsion (Daniel's reactor + experimental docs, 2025-2026)
- Related: PAPER_646 (Universal Inertial Operator, Caduceus, DPM mechanism), PAPER_872

(proto-element nuclear identity), PAPER_087 (BH thermodynamics), PAPER_1156 (cosmology suite), PAPER_1255 (muonic hydrogen radius).

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